

1) RLC circuits

- a) ① V_c can't change instantaneously
 ② i_L can't change instantaneously
 ③ at $t=0$ voltage source steps to $80V$

b) ① $V_L = L \frac{di_L}{dt}$ $V_L(0^+) = V_{source} = 80V$
 $\frac{di_L(0^+)}{dt} = \frac{V_L(0^+)}{L} = 80 / 2 \cdot 10^{-3} = 40 \text{ mA/s}$
 ② $i_C = C \frac{dV_C}{dt}$ $i_C(0^+) = i_L(0^+) = 32 \text{ mA}$
 $di_L(0^+) / dt = 40 \text{ mA/s}$

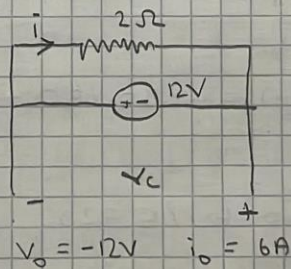
c) $i_R(\infty) = V_{source} / R_1 + R_2 = 80 / 25\Omega + 20\Omega = 1.78 \text{ mA}$

② $i_L(\infty) = i_R(\infty) = 1.78 \text{ mA}$

③ $i_C(\infty) = 0 \text{ A}$

2-)

for $t < 0$,



for $t > 0$,

$$\alpha = R/2L = 2$$

$$\omega_0 = 1/\sqrt{LC} = 2\sqrt{2}$$

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2} = \sqrt{4} = 2$$

$$i(t) = (A \cos 2t + B \sin 2t) e^{-2t}$$

$$i_0 = 6 = A$$

$$di/dt = -2(6 \cos 2t + B \sin 2t) e^{-2t} + (-2 \times 6 \sin 2t + 2B \cos 2t) e^{-2t}$$

$$di(0)/dt = -2 + 2B = (-1/L) [Ri(0) + V_c(0)] = -2 [12 - 12] = 0$$

$$B = 6, \quad i(t) = (6 \cos 2t + 6 \sin 2t) e^{-2t} \text{ A}$$

3-) $V(0^+) = V_0 = 0V$

$$L d^2 v(t) / dt^2 + R dv(t) / dt + v(t) / C = V_s$$

$$3 d^2 v(t) / dt^2 + 46 dv(t) / dt + 27 v(t) = 24 \quad (\text{standard form})$$

$$d^2 v(t) / dt^2 + 46/3 dv(t) / dt + 9 v(t) = 0 \quad (\text{homogeneous form})$$

$$v_h(t) = e^{st}$$

$$3s^2 + 46s + 27 = 0$$

$$s_1 = -0.611$$

$$s = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \rightarrow s = -4.6 \pm 4.233 / 6$$

$$s_2 = -14.055$$

$$v_p(t) \quad 3(0) + 46(0) + 9v = 8$$

$$v_h(t) = A e^{-0.611t} + B e^{-14.055t}$$

$$v(t) = v_h(t) + v_p(t)$$

$$v(t) = A e^{-0.611t} + B e^{-14.055t} + 8/9$$

$$A+B = -8/9$$

$$v(t) = A e^{-0.611t} + B e^{-14.055t} + 8/9$$

$$v(0) = 0$$

4-)

$$t=0, \quad i(0) = 0, \quad v_c(0) = 40 \cdot 30 / 50 = 24V$$

for $t > 0$,

$$\alpha = R / 2L = (40 + 60) / 5 = 20, \quad \omega_0 = 1 / \sqrt{LC} = 1 / \sqrt{2.5 \cdot 10^{-3}} = 20$$

for $\omega_0 = \alpha$,

$$i(t) = [(A+Bt)e^{-20t}], \quad i(0) = 0 = A$$

$$di/dt = ([Be^{-20t}] + [-20(Bt)e^{-20t}]),$$

$$= -(1/2.5) [24] = -9.6$$

$$B = -9.6 \quad \text{or} \quad i(t) = [-9.6 + e^{-20t}] \cdot A$$

5-)

$$\text{for } t = 0, V_o(0) = (81(10)) \cdot 30 = 24 \text{ V}$$

for $t > 0$,

$$\alpha = 1/(2RC) = 1/4$$

$$\omega_0 = 1/\sqrt{LC} = 1/\sqrt{1 \cdot 1/4} = 2$$

since α is less than ω_0 ,

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2} = \sqrt{4 - (1/16)} = 1.9843$$

$$V_o(t) = (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-\alpha t}$$

$$V_o(0) = 30(81(10)) = 24 = A_1,$$

$$i_o(t) = (dV_o/dt) = 0 \text{ when } t=0.$$

$$dV_o/dt = -\alpha (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-\alpha t} + (-\omega_d A_1 \sin \omega_d t + \omega_d A_2 \cos \omega_d t) e^{-\alpha t}$$

$$t=0, \quad dV_o(0)/dt = 0 = -\alpha A_1 + \omega_d A_2$$

$$\text{thus, } A_2 = (\alpha / \omega_d) A_1 = (1/4)(24) / 1.9843 = 3.024$$

$$V_o(t) = (24 \cos(1.9843t) + 3.024 \sin(1.9843t)) e^{-t/4} \text{ V}$$

$$i_o(t) = [0.00031 \cos(1.9843t) - 0.095 \sin(1.9843t)] e^{-t/4} \text{ A}$$

6) a)

$$i_L(0^+) = i_L(0^-) = I_0$$

$$i_L(0^-) = I_0 = 4A \quad i_L(0^+) = 4A$$

$$v_L = L \cdot di/dt \quad v_R = iR \quad v_C = \frac{1}{C} \int i dt$$

RVL $\rightarrow$

$$R - (L \cdot di/dt) - L \cdot di/dt = 0$$

$$di/dt + 8i = 6 \quad i(t) = i_p + i_h$$

$i_h \rightarrow$

$$di/dt = -8t \quad , \quad \ln|i| = -8t + h \rightarrow i_h(t) = h e^{-8t}$$

$i_p \rightarrow$

$$d(i_p)/dt + 8i_p = 6 \rightarrow 8i_p = 6 \quad i_p = 0.75A$$

$$i(t) = i_h + i_p = (h \cdot e^{-8t} + 0.75)A$$

$$\text{at } t=0, \quad i(0) = i_L(0^+) = 4A$$
$$4 = h \cdot e^{-8 \cdot 0} + 0.75$$

$$h = 3.25$$

$i(t)$ for $t > 0$

$$i(t) = 3.25 e^{-8t} + 0.75A$$

$$\tau = \frac{L}{R} = \frac{2}{16} = 0.125s$$

Sinusoidal Steady State

$$1) \quad V_s(t) = 20 \sin(10t - \pi/3) \text{ V}$$

$$i_s(t) = 4 \cos(10t - \pi/4) \text{ A}$$

$$\omega = 10 \text{ rad/s}$$

$$Z_R = R = 10 \Omega, \quad Z_L = j\omega L = j10 \Omega, \quad Z_C = 1/B\omega C = -j5 \Omega$$

$$20 \sin(10t - \pi/3) = 20 \cos(10t - \pi/3 - \pi/2) = 20 \cos(10t - 5\pi/6)$$

$$V_s = 20 \angle -5\pi/6$$

$$i_s(t) = 4 \cos(10t - \pi/4) \quad I_s = 4 \angle -\pi/4$$

$$1/Z_L = 1/Z_R + 1/Z_C \quad Z_L = j10 \quad Z_{RL} = Z_R + Z_L = 10 + j10$$

$$I_{Z_{RL}} = V_s / Z_{total} = \frac{20 \angle -5\pi/6}{10 + j10} \quad |Z_{RL}| = \pi/4$$

$$Z_{total} = 10\sqrt{2} \angle \pi/4 \quad I_{total} = \frac{20 \angle -5\pi/6}{10\sqrt{2} \angle \pi/4} = 1.3 \angle -13\pi/12$$

2)

$$\underline{V_1}$$

$$\frac{V_1}{20} + \frac{V_1 - V_2}{10} + \frac{V_1 - V_2}{-j5} = j2$$

$$(0.15 + j0.2)V_1 - (0.1 + j0.2)V_2 = j2$$

$$\underline{V_2}$$

$$\frac{V_2 - V_1}{10} + \frac{V_2 - V_1}{-j5} + \frac{V_2}{j10} = -(1+j)$$

$$-(0.1 + j0.2)V_1 + (0.1 + j0.1)V_2 = -(1+j)$$

Matrix

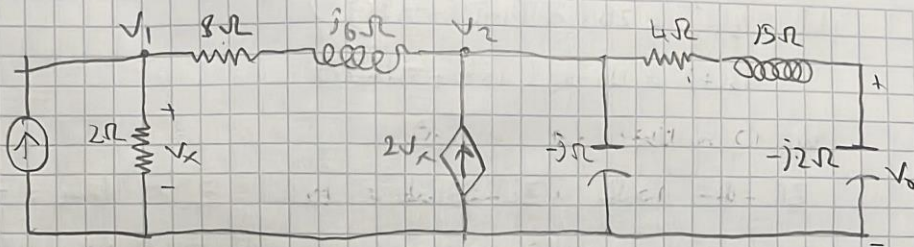
$$\begin{bmatrix} 0.15 + j0.2 & -(0.1 + j0.2) \\ -(0.1 + j0.2) & 0.1 + j0.1 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} j2 \\ -(1+j) \end{bmatrix}$$

$$V = A^{-1} \cdot B$$

$$V_1 = -3.08 - j4.02$$

$$V_2 = -12.31 - j8.46$$

3)



at node 1,

$$4\angle 45^\circ = V_1/2 + (V_1 - V_2)/(8 + j6)$$

$$200\angle 45^\circ = (29 - j3)V_1 - (4 - j3)V_2$$

at node 2,

$$(V_1 - V_2)/(8 + j6) + 2V_x = V_2/-j + V_2/(4 + j5 - j2),$$

$$(104 - j3)V_1 = (12 + j41)V_2$$

$$V_1 = (12 + j41)/(104 - j3)V_2$$

1 plus 2,

$$200\angle 45^\circ = (29 - j3) \cdot (12 + j41)/(104 - j3) - (4 - j3)V_2$$

$$200\angle 45^\circ = (14.21\angle 89.17^\circ)V_2$$

$$V_0 = 5.63\angle 189^\circ V$$

4)

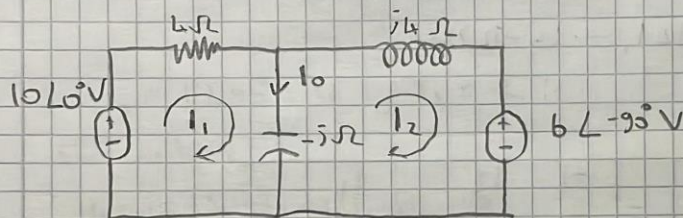
$$\omega = 2$$

$$10 \cos(2t) \rightarrow 10 \angle 0^\circ$$

$$6 \sin(2t) \rightarrow 6 \angle -90^\circ = -j6$$

$$2 \text{ H} \rightarrow j\omega L = j4$$

$$0.25 \text{ F} \rightarrow 1/j\omega C = 1/j(2)(1/4) = -j2$$



For loop 1,

$$-10 + (4 - j2)I_1 + j2I_2 = 0$$

$$5 = (2 - j)I_1 + jI_2$$

For loop 2,

$$j2I_1 + (j4 - j2)I_2 + (-j6) = 0$$

$$I_1 + I_2 = 3$$

matrix,

$$\begin{bmatrix} 2-j & j \\ 1 & 1 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

$$\Delta = 2(1-j), \quad \Delta_1 = 5-j3, \quad \Delta_2 = 1-j3$$

$$i_o(t) = 1.4142 \cos(2t + 45^\circ) \text{ A}$$

5-)

for mesh 1,

$$-40 \angle 30^\circ + (j10 - j20)I_1 + j20I_2 = 0$$

$$4 \angle 30^\circ = -jI_1 + j2I_2$$

for mesh 2,

$$50 \angle 0^\circ + (40 - j20)I_2 + j20I_1 = 0$$

$$5 = -j2I_1 - (4 - j2)I_2$$

for matrix,

$$\begin{bmatrix} 4 \angle 30^\circ \\ 5 \end{bmatrix} = \begin{bmatrix} -j & j2 \\ -j2 & -(4 - j2) \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

$$\Delta = -2 + 4j = 4.472 \angle 116.56^\circ$$

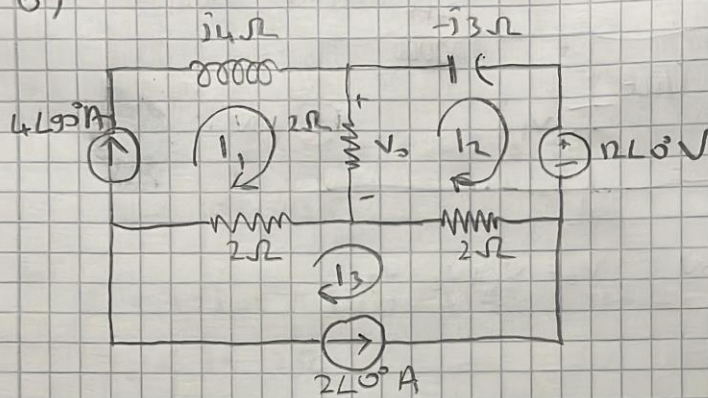
$$\Delta_1 = -(4 \angle 30^\circ)(4 - j2) - j10 = 21.01 \angle 211.8^\circ$$

$$\Delta_2 = -j5 + 8 \angle 120^\circ = 4.44 \angle 156.27^\circ$$

$$I_1 = \frac{\Delta_1}{\Delta} = 4.698 \angle 95.24^\circ \text{ A}$$

$$I_2 = \frac{\Delta_2}{\Delta} = 0.9928 \angle 37.71^\circ \text{ A}$$

6)



$$I_1 = 4\angle 90^\circ = j4, \quad I_3 = -2$$

for mesh 2,

$$(4 - j3)I_2 - 2I_1 - 2I_3 + 12 = 0$$

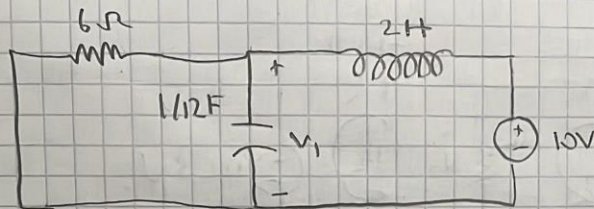
$$(4 - j3)I_2 - j8 + 4 + 12 = 0$$

$$I_2 = (-16 + j8) / (4 - j3) = -3.52 - j0.64$$

thus,

$$V_o = 11.648 \angle 52.82^\circ \text{ V}$$

7)

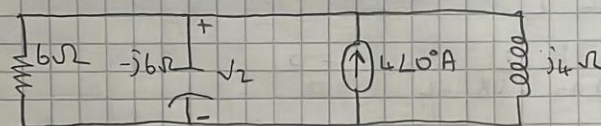


$$V_1 = 10V \quad V_0 = V_1 + V_2 + V_3$$

for V_2 ,

$$\omega = 2, 2H \rightarrow j\omega L = j4$$

$$1/12F \rightarrow 1/j\omega C = 1/(j(2)(1/12)) = -j6$$



$$4 = \frac{V_2}{6} + \frac{V_2}{-j6} + \frac{V_2}{j4} = (1/6 + j/6 - j/4)V_2$$

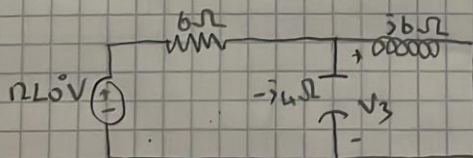
$$V_2 = 24 / (1 - j0.5) = 21.45 \angle 26.56^\circ$$

$$V_2 = 21.45 \sin(2t + 26.56^\circ) V$$

for V_3 ,

$$\omega = 3, 2H \rightarrow j\omega L = j6$$

$$1/12F \rightarrow 1/j\omega C = 1/j3 = -j/3$$



$$(12 - V_3)/6 = V_3 / -j4 + V_3 / j6$$

$$V_3 = 10.73 \angle -26.56^\circ$$

$$V_3 = 10.73 \cos(3t - 26.56^\circ) V$$

$$V_0 = 10 + 21.45 \sin(2t + 26.56^\circ) + 10.73 \cos(3t - 26.56^\circ) V$$

8) sinusoidal V source = $10 \sin(3t - 30^\circ) \text{ V}$

a) Voltage source

$$10 \sin(3t - 30^\circ) = 10 \cos(3t - 30^\circ - 90^\circ) = 10 \cos(3t - 120^\circ)$$

$$V_b = 10 \angle -120^\circ$$

b)

$$Z_C = 1/j\omega C = 1/j3 \cdot (1/6) = -j2 \Omega$$

$$Z_C = 1 - j2 \Omega$$

c)

$$Z_L = j\omega L = j6 \Omega$$

$$I = V/Z \quad I_s = (10 \angle -120^\circ) / (1 - j2)$$

$$|Z| = \sqrt{1^2 + (-2)^2} = \sqrt{5}$$

phase of Z $\rightarrow$

$$\theta_Z = \tan^{-1}(-2/1) = -63.43^\circ$$

$$Z = \sqrt{5} \angle -63.43^\circ$$

$$I_s = 10/\sqrt{5} = 4.47 \quad \angle I_s = -120^\circ + 63.43^\circ = -56.57^\circ$$

$$I_s = 4.47 \angle -56.57^\circ \text{ A}$$

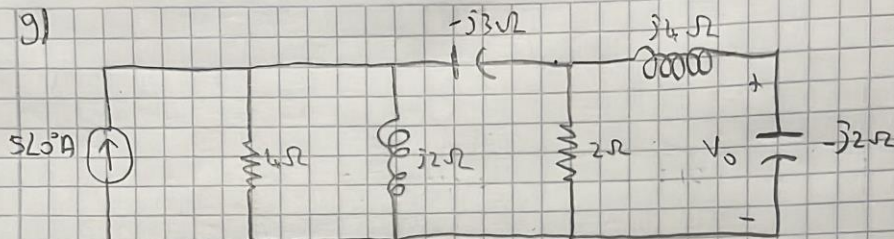
$$I_{s2} = 2 \angle 0^\circ \text{ A}$$

$$I_{\text{total}} = I_s + I_{s2}$$

$$I_{\text{total}} = 5.8 \angle -39.6^\circ \text{ A}$$

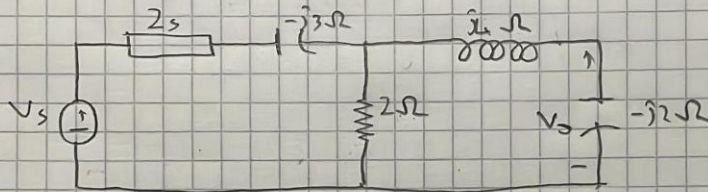
$$i_o(t) = 5.8 \cos(3t - 39.6^\circ) \text{ A}$$

9)



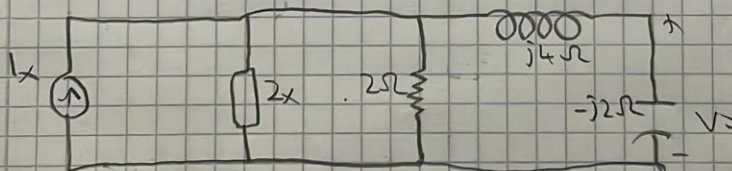
$$Z_s = 4 \parallel j2 = j8 / (4 + j2) = 0.8 + j1.6$$

$$V_s = (5\angle 0^\circ) Z_s = (5)(0.8 + j1.6) = 4 + j8$$

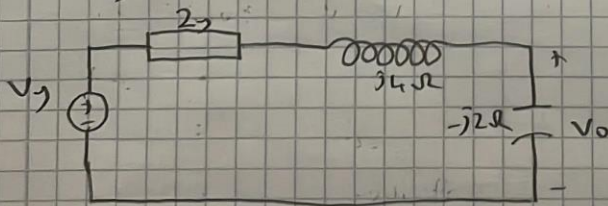


$$Z_x = 2_s - j3 = 0.8 - j1.4$$

$$I_x = V_s / Z_s = (4 + j8) / (0.8 - j1.4) = -3.0769 + j4.6154$$

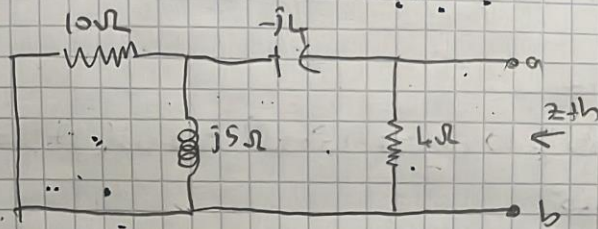


$$Z_y = 2 \parallel 2_x = (1.6 - j2.8) / (2.8 - j1.4) = 0.8571 - j0.5714$$



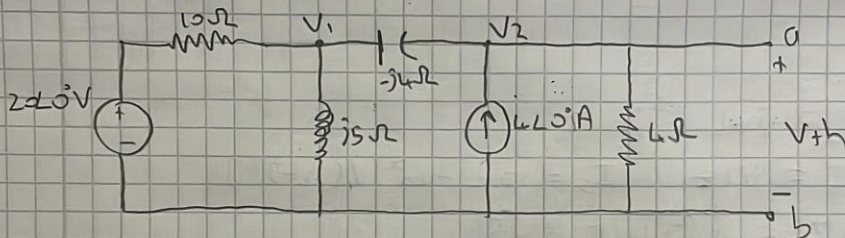
$$V_o = -j2 / (2_y + j4 - j2) V_y = (3.529 - j3.883) V$$

10)



$$Z_{th} = 4 \parallel (-j4 + 10 \parallel j5) = 4 \parallel (-j4 + 2 + j4)$$

$$Z_{th} = 4 \parallel 2 = 1.333 \Omega$$



Node 1,

$$(20 - V_1) / 10 = V_1 / j5 + (V_1 - V_2) / -j4$$

$$(1 + j0.5) V_1 - j2.5 V_2 = 20$$

Node 2,

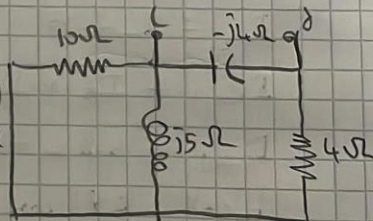
$$4 + (V_1 - V_2) / -j4 = V_2 / 4$$

$$V_1 = (1 - j) V_2 + j4$$

$$V_{th} = V_2 = 9.615 \angle 33.69^\circ \text{ V}$$

$$Z_{th} = -j4 \parallel (4 + 10 \parallel j5) = -j4 \parallel (4 + j10/1)$$

$$Z_{th} = -j4 \parallel (6 + j4) = -j4 \parallel 6(6 + j4) = 2.667 - j4 \Omega$$



$$V_{th} = V_1 - V_2 = 16/3 - j8 = 9.614 \angle 56.31^\circ \text{ V}$$